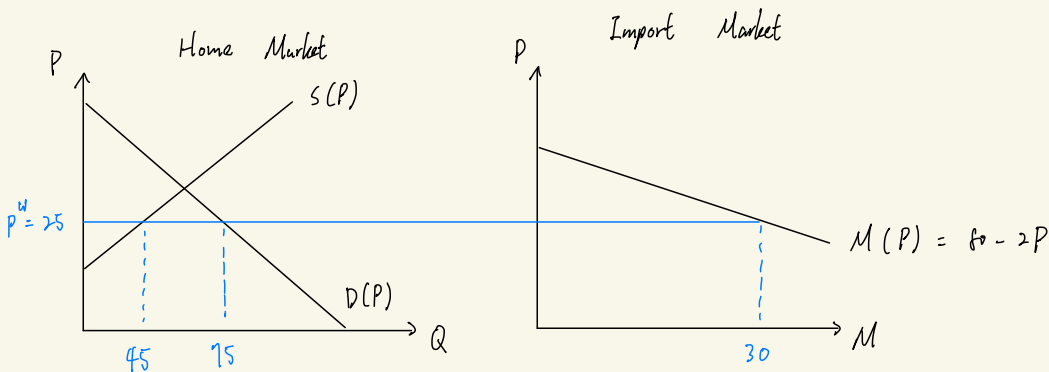


Import tariff : Practice 2

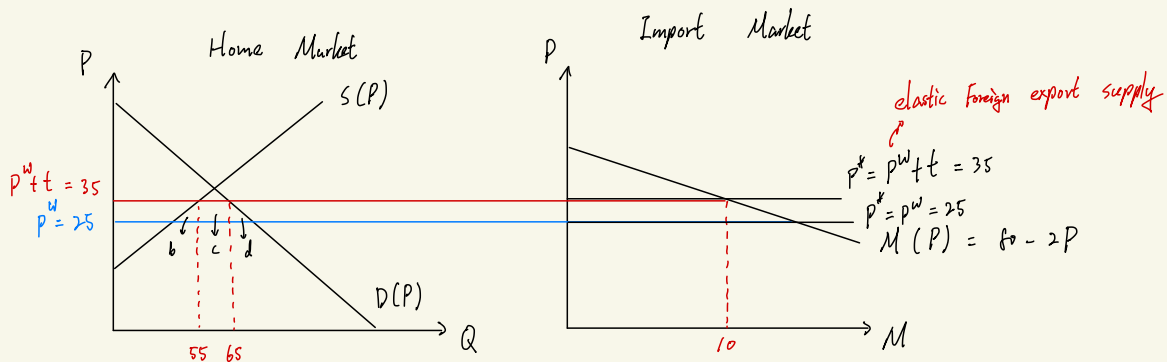
$$D(P) = 100 - P, \quad S(P) = 20 + P, \quad P^W = 25$$

$$M_i = D(P^W) - S(P^W), \quad t = 10$$



1. Home is a small country $\Rightarrow$ elastic foreign export supply

$$P^* = P^W = 25$$



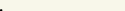
Home price : $P^W + t = 35$

Domestic consumption : $D(P^W + t) = 65$

Domestic production : $S(P^W + t) = 55$

Imports after tariff : $M(P^W + t) = 10$

$$\Delta CS = \text{triangle} - \text{triangle} = \text{trapezoid}$$

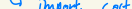
$\Delta PS =$  $-$  $=$ 

$$TR = \begin{matrix} 35 \\ 25 \end{matrix} \begin{matrix} \boxed{} \\ \end{matrix} \begin{matrix} \\ 65 \end{matrix} = 10 \times 10 = 100$$

Transfer : Home consumers -700

- Home suppliers $+500$
- Home government $+100$

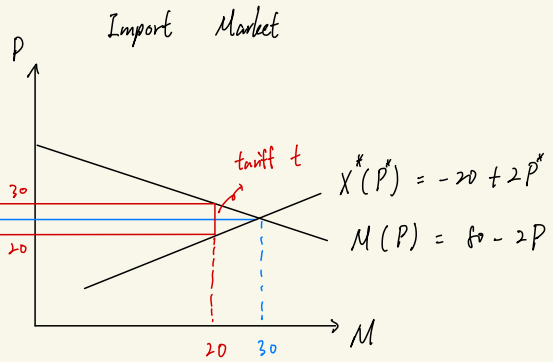
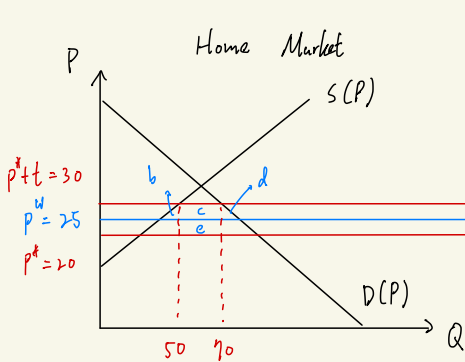
b:  = $\frac{(55-45) \times 10}{2} = 50$

d: 
$$= \frac{(75-65) \times 10}{2} = 50$$

$$\Rightarrow b + d = 100$$

3. Home is a large country $\Rightarrow$ inelastic Foreign export supply

$$X^*(P^*) = -20 + 2P^*$$



Tariff creates the price wedge : $P - P^* = 10$

$\downarrow$
 Home consumers pay
 $\downarrow$
 Foreign suppliers receive

Market clearing condition : $M(P) = X^*(P^*)$

$$\Rightarrow 80 - 2(10 + P^*) = -20 + 2P^* \Rightarrow P^* = 20 \text{ \& } M = 20$$

$$4. \text{ TR} = \begin{matrix} \boxed{\begin{matrix} c \\ e \end{matrix}} \begin{matrix} P \\ P^* \end{matrix} \\ 50 \quad 70 \end{matrix} = 10 \times (70 - 50) = \underline{200}$$

TR here is higher than that in a small country due to "e".

$$b : \triangle = \frac{(50 - 45) \times (30 - 25)}{2} = 12.5$$

$$d : \triangle = \frac{(75 - 70) \times (30 - 25)}{2} = 12.5$$

$$\left. \begin{matrix} b \\ d \end{matrix} \right\} b + d = \underline{25}$$

Total distortion becomes smaller

$$\Delta W_H = \underline{e} - (b + d) = 15$$

term of trade gain : Home pays Foreign less.